

Javier Castro

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Research Interests

My research lies at the interface between scientific machine learning and the numerical analysis of PDEs. I develop mathematically principled neural network methods for nonlinear PDEs, including thermodynamically informed neural networks and weak-residual formulations. In particular, I study how underlying stochastic particle systems can inform the design of structure-preserving loss functionals. As a complementary direction, I study curvature-aware optimization methods for the training of these methods, including Gauss-Newton and natural-gradient methods. A further line of my work concerns the continuous-time weak approximation of stochastic gradient dynamics, and weak approximation methods for SDEs.

Education

Technische Universität Berlin <i>PhD in Mathematics, member of the ERC project FluCo</i> Advisor: Prof. Benjamin Gess. Expected completion: March 2027.	2022-2027 Berlin, Germany
Universidad de Chile <i>MSc in Applied Mathematics</i> Advisor: Prof. Claudio Muñoz. Thesis: <i>Description of local and nonlocal equations using Deep Learning techniques.</i>	2020-2022 Santiago, Chile

Publications

- Javier Castro, Claudio Muñoz, Nicolás Valenzuela.** *The Calderón's Problem via DeepONets.* Vietnam Journal of Mathematics, 52, 775–806, 2024. [link](#).
- Javier Castro.** *The Kolmogorov Infinite Dimensional Equation in a Hilbert Space via Deep Learning Methods.* Journal of Mathematical Analysis and Applications, 527(2), Article 127413, 2023. [link](#).
- Javier Castro.** *Deep Learning Schemes for Parabolic Nonlocal Integro-Differential Equations.* Partial Differential Equations and Applications, 3, Article 77, 2022. [link](#).

Preprints

- Javier Castro, Benjamin Gess.** *THINNs: Thermodynamically Informed Neural Networks.* arXiv preprint, 2025. [link](#).

Selected Talks

THINNs: Thermodynamically Informed Neural Networks SIAM Conference on Optimization, Edinburgh, United Kingdom	2026
The Calderón's Problem via DeepONets PDE Seminar, Universidad de Chile, Santiago, Chile	2023
The Kolmogorov Infinite Dimensional Equation via Deep Learning Methods Jornadas Matemáticas de la Zona Sur, Universidad de Los Lagos, Chile	2022
Deep Learning Schemes for Parabolic Nonlocal Integro-Differential Equations Jornadas Matemáticas de la Zona Sur, Universidad de La Frontera, Chile	2021

Selected Posters

THINNs: Thermodynamically Informed Neural Networks Foundations of Computational Mathematics (FoCM), Vienna, Austria	2026
THINNs: Thermodynamically Informed Neural Networks	

Awards

Jorge Billeke Award for Academic Excellence in Mathematics

2023

Chilean Mathematical Society (SOMACHI) — National distinction for outstanding academic achievement in Mathematics and Mathematical Engineering (for my Master thesis)